THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS CHAPTER TEST SET B

Chapter 11: **Areas Related to Circles** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Use $\pi = \frac{22}{7}$ unless stated otherwise.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The area of a sector of a circle of radius 21 cm and central angle 120° is: 1

(a) 462 cm² **(b)** 231 cm² **(c)** 924 cm² **(d)** 154 cm²

2. The length of an arc of a circle of radius 7 cm subtending 90° is: 1

(a) 11 cm **(b)** 22 cm **(c)** 44 cm **(d)** 5.5 cm

3. If the circumference of a circle is 44 cm, the area of its quadrant is: 1

(a) 38.5 cm² **(b)** 154 cm² **(c)** 77 cm² **(d)** 19.25 cm²

4. The angle swept by the minute hand in 20 minutes is: 1

(a) 120° **(b)** 100° **(c)** 60° **(d)** 20°

5. A student states: "If the radius of a sector is doubled keeping the angle same, its area doubles." The correct response is: 1

(a) Correct **(b)** Incorrect — the area becomes 4 times
(c) The area is unchanged **(d)** The area halves

6. The length of an arc subtending 45° in a circle of radius 28 cm is: 1

(a) 22 cm **(b)** 44 cm **(c)** 11 cm **(d)** 88 cm

7. For a sector of radius 14 cm with arc length 22 cm, Student A says area $= \frac{1}{2} \times 22 \times 14 = 154$ cm²; Student B says area $= 22 \times 14 = 308$ cm². Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

8. The perimeter of a sector of radius r and angle θ is: 1

(a) $2r + \frac{\theta}{360^\circ} \times 2\pi r$ **(b)** $\frac{\theta}{360^\circ} \times 2\pi r$ **(c)** $r + \frac{\theta}{360^\circ} \times \pi r^2$ **(d)** $2\pi r$

9. The area of a semicircle of radius 7 cm is: 1

(a) 77 cm² **(b)** 154 cm² **(c)** 38.5 cm² **(d)** 22 cm²

10. If the area of a sector of a circle of radius 6 cm is 6π cm², its angle is: 1

(a) 30° **(b)** 45° **(c)** 60° **(d)** 90°

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the area of a sector of a circle of radius 10.5 cm with central angle 60°. 2

12. Find the angle subtended at the centre of a circle of radius 7 cm by an arc of length 11 cm. 2

13. A horse is tied to a peg at one corner of a square field of side 20 m by a 7 m long rope. Find the area of the field in which the horse can graze. 2

14. A student says that the area of the minor segment of a circle of radius 7 cm with central angle 90° is 38.5 cm². Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the area of the minor segment of a circle of radius 14 cm when the angle of the corresponding sector is 60°. (Use $\sqrt{3} = 1.73$) 3

16. A sector-shaped flower bed has radius 21 m and central angle 40°. Find 3
(i) its area
(ii) the length of fencing needed along its whole boundary.

17. A chord of a circle of radius 21 cm subtends a right angle at the centre. Find the areas of the minor and major segments. 3

OR

The area of a sector of a circle of radius 14 cm is 154 cm². Find the central angle and the length of the arc. 3

18. The hour hand of a clock is 6 cm long. Find the area swept by it between 11:00 a.m. and 12:30 p.m. Also find the distance moved by its tip. 3

OR

A circular park of radius 28 m has a sector-shaped flower bed with angle 45°. Find the area of the bed and the cost of turfing it at ₹4 per m². 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. In a circle of radius 14 cm, a chord PQ subtends 120° at the centre. Find 5
(i) the length of arc PQ
(ii) the area of the minor sector
(iii) the area of the minor segment
(iv) the area of the major segment. (Use $\sqrt{3} = 1.73$)

OR

Case study: A hand fan is in the shape of a sector of radius 28 cm and angle 135°. 5
(i) Find the area of the fan.
(ii) Find the length of lace needed along its curved edge.
(iii) Find the total cost if lace costs ₹2 per cm and cloth costs ₹0.50 per cm².

20. Case study: A circular cake of radius 21 cm is cut into 6 equal sector-shaped pieces. 5
(i) Find the angle of each piece.
(ii) Find the area of the top of each piece.
(iii) Find the length of the curved crust of each piece.
(iv) A student says 2 pieces together cover 462 cm². Verify.

OR

A sector of angle 150° is cut from a circular sheet of radius 21 cm. Find 5
(i) the length of the arc
(ii) the area of the sector
(iii) the area of the sheet left.

END OF QUESTION PAPER — **Total: 40 Marks**